
OE-EC803A: Internet of Things (IoT)

Quick-Recall One-Liners Revision Guide

Semester VIII B.Tech ECE (MAKAUT) • Prepared by Firdous Rahaman • June 12, 2026

Chapter 1: Introduction to IoT

1. **Internet of Things (IoT) definition** → Global network of self-configuring physical and virtual objects.
2. **World Wide Web vs. IoT core distinction** → WWW connects humans; IoT connects physical and virtual devices.
3. **Enchanted Object definition** → Everyday physical object enhanced with ambient connectivity and sensors.
4. **Localized network focused purely on data collection** → Wireless Sensor Network (WSN).
5. **Traditional Machine-to-Machine (M2M) communication style** → Closed, point-to-point, isolated, and hardware-centric.
6. **Direct local device communication model** → Device-to-Device (D2D).
7. **Direct internet platform connection model** → Device-to-Cloud (D2C).
8. **IoT Gateway primary function** → Performs protocol translation and data routing.
9. **ETSI M2M architecture interfaces** → dIa (device-app), mIa (network-app), mId (M2M-device).
10. **Generic M2M system solution layers** → M2M Area Network, Communication Network, M2M Applications.
11. **Four stages of IoT information value chain** → Sense, Transmit, Store/Analyze, Decide/Act.
12. **OGC API standard for querying sensor observations** → SOS (Sensor Observation Service).
13. **OGC XML schema for sensor parameters** → SensorML (Sensor Model Language).
14. **Category used for business-to-consumer processes** → Personal IoT.
15. **Category used by citizens to benefit smart cities** → Community IoT.
16. **Automatic device communication without human intervention** → Machine-to-Machine (M2M).
17. **Network of uniquely identifiable embedded computing devices** → Internet of Things (IoT).
18. **Core components providing physical world awareness** → Sensors.
19. **Vital elements required by house key computers** → Data and services.
20. **M2M device domain alternative name** → M2M Area Network.
21. **Industrial IoT (IIoT) priority focus** → Zero-downtime, safety, and mission-critical reliability.

Chapter 2: Design Principles for Connected Devices

1. **Calm Technology primary goal** → Blends into the periphery, reducing cognitive fatigue.
2. **Calm Technology design rules** → Peripheral attention, seamless shift, presence, reach, attention limits.
3. **Ambient Technology definition** → Devices displaying status information via subtle environmental cues.
4. **Ambient Orb designer** → David Rose.
5. **Ambient Orb feedback mechanism** → Frosted sphere changing color using internal LEDs.
6. **Ambient Orb main limitation** → Low information density (conveys status, not text).
7. **Magic as Metaphor concept** → Using mythical archetypes to explain wireless operations.
8. **Gesture-controlled magical controller device example** → Kymera Magic Wand.
9. **Critiques of Magic as Metaphor** → Expectation mismatch, lack of debugging/error states.
10. **Privacy design solution for local devices** → Edge processing and data minimization.
11. **Web Thinking for Connected Devices definition** → Building IoT architectures using standardized HTTP/REST principles.
12. **URI role in Web Thinking** → Unique address identifier for every physical sensor/actuator.
13. **REST command to read sensor state** → GET.
14. **REST command to trigger actuator action** → POST.
15. **Physical Mashup definition** → Linking different physical devices using standard web APIs.
16. **Affordance definition** → Physical properties of an object suggesting its usage.
17. **Clash of Affordances in IoT** → Traditional appearance hiding invisible digital sensor interfaces.
18. **Visual signifiers resolving Clash of Affordances** → LED rings or textured borders guiding interaction.
19. **Graceful degradation design requirement** → Device remains functional physically if internet or power fails.
20. **Ambient device pulsing when rain is expected** → Ambient Umbrella.

Chapter 3: Internet Principles

1. **BLE full form** → Bluetooth Low Energy.
 2. **BLE GATT hierarchy data structures** → Profiles, Services, and Characteristics.
 3. **Classic Bluetooth vs. BLE latency** → Classic up to 6s; BLE under 3ms.
 4. **Address style permanent and manually configured** → Static IP.
 5. **DHCP dynamic configuration handshake acronym** → DORA.
 6. **DORA handshake steps** → Discover, Offer, Request, Acknowledge.
 7. **TCP full form** → Transmission Control Protocol.
 8. **UDP full form** → User Datagram Protocol.
 9. **Reliable, connection-oriented transport protocol** → TCP.
 10. **Fast, lightweight, connectionless transport protocol** → UDP.
 11. **UDP minimum header size** → 8 bytes.
-

12. **TCP minimum header size** → 20 bytes.
13. **IPv6 address length in bits** → 128 bits.
14. **IPv4 address length in bits** → 32 bits.
15. **IPv6 feature for stateless autoconfiguration** → SLAAC.
16. **MAC address length in bits** → 48 bits.
17. **MAC address data link layer level** → Layer 2 (Physical address).
18. **MAC address manufacturer identifier first 24 bits** → OUI (Organizationally Unique Identifier).
19. **Logical endpoints routing network traffic in OS** → Network Ports (16-bit).
20. **HTTP message format style** → ASCII text headers.
21. **CoAP transport protocol and minimum header size** → UDP; 4 bytes.
22. **MQTT architecture model and minimum header size** → Publish-Subscribe (Broker-based); 2 bytes.
23. **Weightless protocol radio spectrum utilization** → Sub-GHz TV White Space (TVWS).
24. **Computing proximity closest to data source** → Mist computing.
25. **Proximity order of computing layers** → Mist → Edge → Fog → Cloud.
26. **Computing layer running on regional routers** → Fog computing.
27. **Layer connecting IoT devices with WAN** → Network layer.
28. **IoTA full form** → Internet of Things Architecture.
29. **Protocol linking all devices in the internet** → TCP/IP.
30. **Computing emphasizing proximity to end user** → Fog computing.

Chapter 4: Thinking About Prototyping

1. **Prototyping sketching purpose** → Fast, low-cost exploration of concepts.
2. **Low fidelity prototype main benefit** → High ease of creation and low initial cost.
3. **High fidelity prototype main benefit** → Evaluates real-time performance and user experience.
4. **Sourcing model prioritizing IP protection and restricted designs** → Closed Source.
5. **Sourcing model bypassing royalty fees and accelerating development** → Open Source.
6. **MVP full form in startup prototyping** → Minimum Viable Product.
7. **Scale-up design challenge in prototyping** → Transitioning from breadboards to mass-manufactured PCBs.
8. **Prototyping board with breadboard friendly pins** → Arduino Uno.
9. **Community-driven hardware source code repository** → GitHub.
10. **Value of tapping into the community** → Reusable libraries, fast debugging, and hardware templates.

Chapter 5: Prototyping Physical Design

1. **3D printing technology style** → Additive manufacturing.
2. **Filament material standard for FDM 3D printing** → PLA (Polylactic Acid) or ABS.

3. **Laser cutting process style** → Subtractive manufacturing.
4. **Laser cutter material compatibility constraint** → Dangerous fumes from PVC/plastics containing chlorine.
5. **CNC milling definition** → Computer-controlled rotating tool cutting block material.
6. **Subtractive method for high-accuracy metal brackets** → CNC milling.
7. **Fast method for enclosure mockups** → Laser cut cardboard or acrylic.
8. **3D printing technique using UV laser on resin** → SLA (Stereolithography).
9. **Recycling/repurposing in physical prototyping** → Reusing existing enclosures or containers.
10. **File format for 3D printing** → STL (Stereolithography) or 3MF.

Chapter 6: Prototyping Embedded Devices

1. **Arduino processor architecture style** → AVR/ARM 8-bit or 32-bit Microcontroller.
2. **Raspberry Pi processor architecture style** → ARM 64-bit Microprocessor.
3. **Arduino boot speed characteristics** → Instant boot.
4. **Raspberry Pi boot time** → 15 to 30 seconds.
5. **Arduino standard RAM capacity** → 2 KB SRAM (ATmega328P).
6. **Raspberry Pi RAM capacity range** → 1 GB to 8 GB LPDDR4.
7. **GPIO full form** → General Purpose Input Output.
8. **I2C serial bus lines** → SDA (Data) and SCL (Clock).
9. **SPI serial bus lines** → MOSI, MISO, SCLK, and CS (Chip Select).
10. **UART serial bus lines** → TX (Transmit) and RX (Receive).
11. **Raspberry Pi GPIO logic voltage limit** → 3.3V.
12. **BeagleBone Black ADC resolution** → 12-bit (7 analog input channels).
13. **BeagleBone co-processors for deterministic real-time I/O** → PRUs (Programmable Real-Time Units).
14. **Electric Imp split coding architecture** → Device code and Cloud code.
15. **Electric Imp programming language** → Squirrel.
16. **Electric Imp vendor constraint** → Closed vendor lock-in.
17. **Smartphone prototyping advantage** → Rich built-in sensors and wireless communication.
18. **Plug Computer primary advantage** → Always-on mains powered 24/7 server design.
19. **Arduino programming language wrapper base** → C/C++ wrapper.
20. **PCI Express role in IoT gateways** → Interconnecting high-speed Wi-Fi, Bluetooth, or cellular cards.
21. **Ethernet 10/100M support connection target** → Local Area Network (LAN).
22. **More powerful model between Pi 2 and Pi 3** → Raspberry Pi 3.

Chapter 7: Prototyping Online Components

1. **API full form** → Application Programming Interface.
 2. **REST API resource formats** → JSON or XML.
-

3. **API architecture consideration beyond tech** → Security, access control, and authentication.
4. **Protocol for real-time bidirectional web communications** → WebSockets.
5. **MQTT broker role in online components** → Directs publisher messages to target subscribers.
6. **Push notification protocol for mobile alerts** → APNs (Apple) or FCM (Firebase).
7. **Standard auth method for securing APIs** → OAuth 2.0 / API Keys.
8. **CORS full form in API development** → Cross-Origin Resource Sharing.
9. **HTTP code for resource not found** → 404.
10. **HTTP code for successful request** → 200 OK.
11. **Scope of API architecture beyond technical elements** → Multi homing (mobile networks with dynamic membership).

Chapter 8: Techniques for Writing Embedded Code

1. **Primary risk of dynamic memory allocation in MCU** → Heap fragmentation leading to crashes.
2. **Dynamic memory allocation alternative** → Static allocation of fixed-size buffers.
3. **Blocking delay side effect** → Halts CPU cycles, blocking event handlers.
4. **Non-blocking delay implementation** → State machines using 'millis()' checks.
5. **Battery life optimization technique using low power** → CPU deep sleep modes.
6. **Action to audit third-party libraries** → Checking Flash and RAM consumption reports.
7. **Serial print debugging drawback** → Alters timing execution and consumes RAM/UART.
8. **JTAG/SWD debugging advantage** → Non-intrusive breakpoints and real-time register audit.
9. **Reliability phase forwarding packets hop-by-hop** → Message relaying.
10. **Reliability phase tracking sequence counter gaps** → Lost message detection.
11. **Reliability phase asking for missing packets only** → Selective recovery.
12. **Reliability first step in WSN reliable transfer** → Initialization.
13. **CPU clock reduction scaling method** → DVFS (Dynamic Voltage and Frequency Scaling).
14. **Non-intrusive debugging probe connector** → JTAG or SWD.
15. **Best storage for persistent configuration settings** → EEPROM or Flash.

Chapter 9: Prototype to Reality (Business Models & Manufacturing)

1. **Business Model Canvas creator** → Alexander Osterwalder.
2. **Number of building blocks in BMC** → 9.
3. **BMC block defining product value to customers** → Value Proposition.
4. **Lean Startup build-measure-learn cycle first stage** → Build a Minimum Viable Product (MVP).
5. **Core decision after build-measure-learn evaluation** → Pivot or Persevere.
6. **Startup funding using small public contributions** → Crowdfunding (Kickstarter).

7. **PCB full form** → Printed Circuit Board.
8. **First step in PCB design workflow** → Creating the schematic diagram.
9. **Manufacturing technique for high-volume plastic enclosures** → Injection molding.
10. **Certification required for electronic sales in Europe** → CE.
11. **Certification required for radio devices in USA** → FCC.
12. **Startup funding from high-net-worth individuals** → Angel Investment.

Chapter 10: Ethics & Smart Cities

1. **Smart City definition** → Urban area optimizing utility services using IoT data.
2. **Smart Lighting energy savings percentage** → Up to 80% using ambient LDR sensors.
3. **Smart Waste sensor type** → Ultrasonic distance sensor.
4. **Smart waste routing optimization** → Skips empty bins, planning routes dynamically.
5. **V2X full form in Intelligent Transportation Systems** → Vehicle-to-Everything.
6. **Priority route technique for emergency vehicles** → Emergency traffic preemption.
7. **Smart City baseline data ingestion layer** → Integrated Information Provider.
8. **Smart City command and control center** → Management Center.
9. **Smart City citizen interface touchpoint layer** → Mobile Unified Service.
10. **Smart City GIS analytics layer** → Urban Application Platform.
11. **Passive constant monitoring of public without consent** → Surveillance Creep.
12. **Deducing personal habits from benign appliance logs** → Data Profiling.
13. **Remote bricking definition** → Disabling device functionality remotely by vendor.
14. **Botnet created by compromising default password IoT** → Mirai botnet.
15. **Mitigating vendor lock-in using standardized APIs** → Open standards adoption (Matter).
16. **Stake level in physical security** → High.
17. **Reliable bidirectional signaling challenge category** → Signaling.
18. **Availability of video/data hosting** → Security system.
19. **IoT service manageability feature** → Simple and fast installation.
20. **Ethics pillar for modular/recyclable materials** → Circular design / environmental impact.